Stacking Galaxies Reveals a Universal Baryon Fountain

Averaging many simulated galaxies exposes regular, coherent gas flows from disk to halo and back.

arXiv:2609.21374 — Stacked baryon cycles in cosmological hydrodynamic simulations

Authors: Chengyu Ma, Yangyao Chen, Enci Wang, Houjun Mo, Huiyuan Wang, Tao Wang, Qirong Yuan, Min Du et al.

arXiv:2609.21374 doi:10.48550/arXiv.2609.21374

Background

Galaxies grow by cycling gas inward from their surroundings and expelling it through feedback. These fountain-like flows are faint and messy in any single system.

> **Individual galaxy ecosystems are too diverse and dynamic for the underlying baryon-cycle pattern to emerge clearly.**

What this paper is selling

The authors stack gas velocity and property fields across ensembles of simulated galaxies, suppressing scatter to reveal a strikingly regular fountain. The pattern matches observations and admits a compact parameterization.

Key results to remember

- Stacking yields smooth, coherent inflow and outflow cones across the full ecosystem extent.
- The fountain geometry persists across stellar mass and redshift, expanding with halo size.
- A simple parameterized model reproduces the stacked velocity field with small residuals.

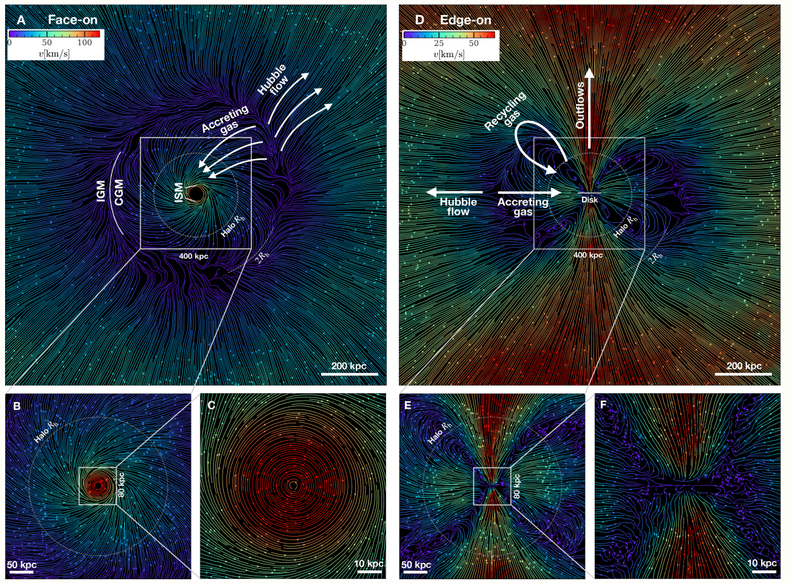


Figure 1 | Stacked streamlines reveal a smooth, coherent gas velocity field from disk plane out to the virial radius.

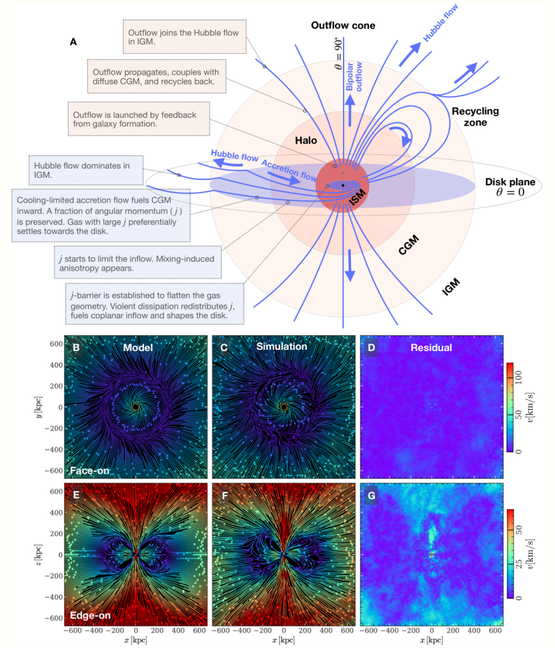


Figure 4 | A schematic fountain model, tuned to the simulation, reproduces the stacked velocity field with small residuals.